Reading Plan — The Realization-Theorem Technique

(the σ-essential fourth-cell route)

Patrick S. Mahon
Personal reference

July 2026

Why this plan exists

This is a *technique-acquisition* plan, companion to the broader `sigma_essential_skill_plan` (which sequences descriptive set theory and large cardinals). It is narrower and pointed at one route.

The σ-essential import sweep closed every *existing* object one could borrow as the carrier's skeleton: each dies either Boolean-ambient (rescued by a non-Dirac σ-state) or operator-algebraic (non-concrete by Kochen–Specker). The one surviving *lead* — not a solution, a place to learn — is the family of **realization theorems**. Per the prior-art note (`czech_school_prior_art_sigma_essential.md` §POSITIVE), these build *non-simplex* (hence contextual, non-segregated) state spaces on orthomodular lattices by *pasting Boolean blocks* — a combinatorial, non-operator-algebraic construction, so the orthocomplement is block-inherited and the route dodges the polarity/form horn that kills matroids and geometric lattices.

The one crux — read every source against it. The realization theorems in the literature are *finite-pasting* constructions and realize the *full (finitely additive)* state space. They fail the σ-essential target on a single **conjunction gap**:

> *Why must the pasting be finite, and can an infinite pasting realize a non-simplex state space on an OML that is* ***simultaneously*** *concrete and σ-complete?*

No located paper does this (prior-art note, §POSITIVE). That conjunction — concrete + σ-complete + non-simplex-surviving-restriction-to-σ-states — is exactly the open cell. Do not read these sources for "realization theory" in general; read each only for how it bears on the conjunction gap.

Citation caveat. The realization papers below are **not yet in the repo bib**; venue/year/page are reconstructed from the prior-art note and are marked [cite unverified]. Verify against the primary source before relying on any detail. Claims about paper *content* here are the prior-art note's summary, not a reading of the papers themselves.

What is in the repo library (checked 2026-07)

- **Harding 2004, "Remarks on Concrete Orthomodular Lattices"** (`harding_2004_concrete_oml.pdf`, 20pp) — **IN HAND; read this first.** Primary source on Kalmbach's $K(L)$ and the concreteness half. Read (this session) confirms: concrete $\equiv$ full set of two-valued states (Godowski 1981, = our C1 exactly); $K(L)$ "glues the Boolean algebras generated by the chains of L" and *every* $K(L)$ *is concrete* (Mayet–Navara 1995) with orthocomplement = set-complement, **block-inherited, not form-derived** — so the polarity/form horn is dodged, confirmed from a primary source. **But** $K(L)$ **is finitary** (finite orthogonal joins; no σ-completeness): this *is* the finiteness dependency the plan hunts, now located. The paper's own reframe: $K(L)$

forms *do not generate* the variety of all concrete OMLs, so a concrete σ-witness **need not be a paste at all** (see the impossibility-twin caveat below).

- **Navara 1992, "Regularity and σ-additivity of states"** (`navara_1992.pdf`, 3pp) — IN HAND; step-4-adjacent (σ-additivity of states, the regularity kill).
- **Müller 1993, "Jauch–Piron States on Concrete Quantum Logics"** (`muller_1993.pdf`) — IN HAND; a concrete non-Boolean example (JP-focused, not a σ-essential witness; useful as a concrete-logic specimen).
- **Pták–Pulmannová 1994** (`ptak_pulmannova_1994.pdf`, 5pp) is the *Floor paper* (subadditive measure ⇒ Boolean), **not** the textbook.
- **To fetch: Kalmbach, *Orthomodular Lattices*** (1983) — the textbook for the pasting mechanics (§1 below); Navara–Rogalewicz, Harding–Navara, Mayet–Navara 1995, Navara–Rüttimann 1991 (§§2–4 below).

1 Foundations: how block-pasting builds an OML

Kalmbach, *Orthomodular Lattices* (1983) [to fetch], with Harding 2004 (in hand) as the concrete-side entry point, and Pták–Pulmannová *Orthomodular Structures as Quantum Logics* [**?**] [the book, to fetch — distinct from the 5pp Floor paper in the library] for the concrete-logic side. The pasting construction and horizontal sums.

- State the pasting of a family of Boolean blocks precisely: how blocks overlap, what the resulting order and orthocomplement are. *Confirm* the orthocomplement is inherited blockwise, not derived from any form — this is why the route dodges the scout's polarity horn.
- Where does the loop / commutativity condition force the paste to be an OMP rather than a lattice? (This is one place "infinite non-segregated pasting" can silently fail before σ even enters.)
- Contrast a *segregated* paste (horizontal sum, contextuality on a central factor — $\prod_n MO_2$, MO_κ) with a non-segregated one. What structural feature makes the incompatibility non-central? The target needs the latter.

2 The realization theorems (read for: where finiteness enters)

Navara–Rogalewicz, "The pasting constructions for orthomodular posets," Demonstratio Math. (1988) [cite unverified]. The base realization: every (compact convex) set arises as a state space of a pasted OMP.

- Read for the *exact step* where finiteness of the pasting is used to realize a non-simplex state space. Which part of the argument breaks if the family of blocks — or an individual block — is infinite?
- The realized object is the *full, finitely additive* state space. Is there any control on σ-additive states? (Prior-art note: no — this is the unaddressed point.)

Harding–Navara, "Subalgebras of orthomodular lattices," Order 17 (2000) [cite unverified]. The strengthened realization: prescribed center and automorphism group.

- "Prescribed center" is the lever for *non-segregation*: a trivial (or small) center forces contextuality off any central Boolean factor. Read for how the center is controlled, and whether that control survives an infinite paste.
- Same finiteness question as above: locate the finite-pasting dependency.

3 The concreteness half (read for: what buys set-representability)

Mayet–Navara (1995) [cite unverified]. Concreteness for pasted OMLs holds only for special forms (prior-art note names $K(L)$).

- What is the special form that yields a set-representable (concrete) paste, and is it compatible with the *infinite* pasting the σ-side needs? Concrete + σ together is the whole gap; this is the paper on the concreteness half.
- Concreteness = enough two-valued states to separate points ($a \mapsto \{s : s(a) = 1\}$ is an embedding). Does the $K(L)$ form preserve a separating supply of *two-valued* states, or only states in general?

4 The σ-boundary (read for: what survives restriction to σ-states)

Navara–Rüttimann, "A characterization of σ-state spaces of orthomodular lattices," Expo. Math. 9 (1991) 275–284 [cite unverified; NOT in repo bib — the existing `Ruttimann1994` key is a different Rüttimann paper ("Weakly purely finitely additive measures," Canad. J. Math. 46), do not conflate].

- The theorem places $S_\sigma(P)$ as a *semi-exposed face* of the full state space $S(P)$ (per Pták's zbMATH review). Read for: does a *non-simplex* full state space, restricted to this face, stay non-simplex? A face of a non-simplex set need not be a simplex — so this is the gate the realized non-simplex must pass to remain contextual for σ-additive states.
- This is the precise juncture where the finite-pasting realization would have to be shown to survive σ-restriction. It is the mathematical heart of the push.

5 The impossibility twin (one entry — do not over-invest)

The same σ-state-space characterization (Navara–Rüttimann 1991) is also the tool for the *negative* attempt: if the semi-exposed face is forced to be a simplex with dispersion-free vertices on every concrete σ-complete OML, the cell is empty. The prior-art note records that the verbatim theorem does *not* force this — so the emptiness proof, if it exists, needs more. Read the characterization once for both directions; the new-world and impossibility branches stay indistinguishable until one is worked.

Caveat raising the impossibility bar (Harding 2004). Since $K(L)$ forms do not generate the variety of all concrete OMLs, an impossibility proof **cannot** be won by showing "no *paste* is σ-essential" — non-paste concrete OMLs exist, and any of them could carry the witness. A $\neg\Psi$ argument must range over *all* concrete σ-complete OMLs, not just Kalmbach constructions. (This does not escape the construct-vs-assume fork: a non-paste concrete OML is still cut out by *some* disjoint-union-closure rule on $\mathcal{P}(\Omega)$, to which the fork applies.)

The order

Kalmbach/Pták–Pulmannová (pasting mechanics) $\to$ Navara–Rogalewicz then Harding–Navara (locate the finiteness dependency) $\to$ Mayet–Navara (concreteness form) $\to$ Navara–Rüttimann (the σ-restriction gate). The push, if it works, is a single new theorem: an *infinite*, concrete, σ-complete paste whose non-simplex state space survives restriction to σ-additive states. Everything above is read to find where that theorem must do its work, and where the finite-pasting literature stops.